

# Mobile Application Development (BIT4107) Portfolio

**Author:** Sheila Muriithi

**Admission Number:** BIT/2024/32455

**Course:** BSc Information Technology

## Overview

This repository contains the complete coursework and project files for the BIT4107 unit. The portfolio demonstrates a progressive learning curve in mobile application development, starting from foundational Flutter environment configurations to the implementation of clean-architecture local storage, and culminating in a highly interactive, multi-screen frontend prototype.

## Contents

|                                                   |   |
|---------------------------------------------------|---|
| Assignment 1: Flutter Foundation .....            | 2 |
| Assignment 2: Mini Student Management System..... | 2 |
| Assignment 3: Gathimaini Pharmacy Prototype.....  | 3 |
| Documentation .....                               | 3 |

## Assignment 1: Flutter Foundation

A foundational mobile application built to demonstrate successful environment setup and basic UI manipulation in Flutter.

### Key Features:

- Modified application title to Assignment One.
- Customized the primary background color to a specific Light Teal (0xFFE0F2F1).
- Updated typography, implementing a bold font size of 35.0 for the main text.
- Integrated a functional ElevatedButton that triggers a success SnackBar upon interaction.

### Running the App:

1. Navigate to the assignment\_one directory.
2. Ensure an emulator is running or a device is connected.
3. Execute the command: flutter run.

## Assignment 2: Mini Student Management System

A functional student registry application that simulates local data persistence and utilizes a modular clean-architecture pattern.

### Key Features:

- **Clean Architecture:** Codebase separated into models, mock database, and screen layers.
- **Authentication:** Features a centralized login portal and secure routing.
- **Registration Validation:** Form logic validates against a local in-memory array (mock\_db.dart) to prevent duplicate registration numbers before saving the record.
- **Dynamic Dashboard:** A modern profile interface that dynamically renders the list of active, registered students.

### Running the App:

1. Navigate to the Assignment 2 directory.
2. Run flutter pub get to fetch dependencies.
3. Execute the command: flutter run.

## Assignment 3: Gathimaini Pharmacy Prototype

A comprehensive, mobile-first frontend prototype designed in Figma and engineered using React and modern web technologies. The Figma workspace utilizes fully interactive native layers rather than flat images to ensure editable, high-fidelity design inspection.

### System Features:

- **7 Interactive Screens:** Includes Login, Home Dashboard, Medicine Details, Cart, Prescription Upload, User Profile, and Wireframes.
- **Mobile-First Viewport:** Engineered with a custom PhoneFrame component to simulate a native mobile environment and aspect ratio.
- **Dynamic Cart Logic:** Features a functional cart that manages item quantities, subtotal calculations, and a standardized delivery fee.
- **Bottom Navigation:** Persistent routing implemented via React Router for seamless transitions between major app modules.

### Tech Stack:

- React & TypeScript
- React Router v7
- Tailwind CSS v4
- Motion (Animations)
- Radix UI Components

### Running the Prototype:

1. Navigate to the Assignment 3 project directory.
2. Run `npm install` to install all required dependencies.
3. Run `npm run dev` to launch the local development server.

### Documentation

The official project logbook and technical report are included in the root of this repository.

- **Mobile app Documentation.docx:** Contains weekly progress tracking, environment setup visual proof, terminal outputs, and the final compiled project report.